SPA617

RADIO ASTRONOMY LABORATORY

Experiment 2 : Antenna Characterization

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1 Antenna Characterization

1.1 Aim

To characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

1.2 Objectives

(a) To determine the antenna operating frequency and return loss from VSWR measurements.

(b) To measure the effect of distance on the power received by the antenna.

(c) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(d) To determine the half-power beamwidth and estimate the directivity of the antenna.

1.3 Theory

1.3.1 Basic antenna concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. An antenna can be used either as a transmitting antenna or as a receiving antenna.

A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$), with a feed line connected between the two arms, as shown in Fig. 1. It radiates most strongly in directions perpendicular to the wire and has very little radiation along the wire.

A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength ($\lambda/4$). The ground acts like a mirror, making it behave like a full dipole.

- A transmitting antenna converts electrical signals into electromagnetic waves and radiates them.
- A receiving antenna converts the received electromagnetic waves into electrical signals.
- In two-way communication, the same antenna can be used for both transmission and reception.

1.3.2 Antenna beam pattern

The energy radiated by an antenna is represented by the radiation pattern of the antenna. The radiation pattern generally has a main lobe, side lobes, and a back lobe.

- The main lobe is the region where the maximum radiated energy exists. The direction of this lobe indicates the principal direction of the antenna beam.
- The other parts of the pattern where radiation is distributed sideways are known as side lobes or minor lobes.
- The lobe opposite to the main lobe is known as the back lobe.

E-plane and H-plane angular measurements are used to determine the radiation beam pattern of an antenna. The E-plane is the plane containing the electric-field vector and the propagation vector, while the H-plane is the plane containing the magnetic-field vector and the propagation vector. The E-plane and H-plane are perpendicular to each other.

1.3.3 Reflection coefficient, VSWR, and return loss

When the antenna impedance Z_L differs from the characteristic impedance Z_0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. For a perfectly matched antenna,

$$Z_L = Z_0, \qquad |\Gamma| = 0.$$

The voltage standing-wave ratio is related to the magnitude of the reflection coefficient by

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}.$$

The return loss is defined as

$$RL = -20\log_{10}|\Gamma| \qquad \text{dB}.$$

A well-matched antenna has a VSWR close to unity, a small reflection coefficient, and a large return-loss value. The resonant or operating frequency is generally identified near the minimum of the VSWR curve.

1.3.4 Half-power beamwidth

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the -3 dB points.

- A small HPBW indicates a narrow, focused beam.
- A large HPBW indicates a broad beam.

1.3.5 Directivity

Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. The directivity is related to the beam solid angle Ω_A as

$$D = \frac{4\pi}{\Omega_{A.}}$$

For practical antennas, the beam solid angle can be approximated using the half-power beamwidths in the E-plane (θ_E) and H-plane (θ_H):

$$D \approx \frac{4\pi}{\theta_E \theta_{H,}}$$

where θ_E and θ_H are expressed in radians.

1.4 Components Required

- Monopole and dipole antennas
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser

- RF Tee connector
- 50 Ω termination
- Antenna support stand
- Coaxial cables and suitable RF adapters

1.5 Experimental Setup

Connect the transmitting antenna to the signal generator and the receiving antenna to the spectrum analyser, as shown in Fig. 4. Keep the antennas at the same height and align them appropriately for the measurement being performed.

For the VSWR and return-loss measurement, connect the signal generator, RF Tee connector, antenna (DUT), and oscilloscope as shown in Fig 5. The oscilloscope measures the variation produced by the incident and reflected signals at the Tee junction.

1.6 Experimental Procedures

1.6.1 Part A.1: Antenna operating frequency and Far-Field Boundary :

Compare the measured operating frequency with the theoretical resonance frequency of a $\lambda/4$ monopole or a $\lambda/2$ dipole, as applicable.

$\lambda/4 = 0.05\text{cm} \rightarrow \lambda = 0.2\text{m}$

$f_{operating} = c/\lambda = 1.5\text{GHz}$

To ensure proper beam pattern measurements , the minimum far-field distance was verified using the maximum aperture dimension (D=0.29m) and the signal generator power set to 20dBm :

$R_{far\text{-}field} = 0.841\text{m}$

Thus, all angular measurements are taken beyond this 84.1cm boundary.

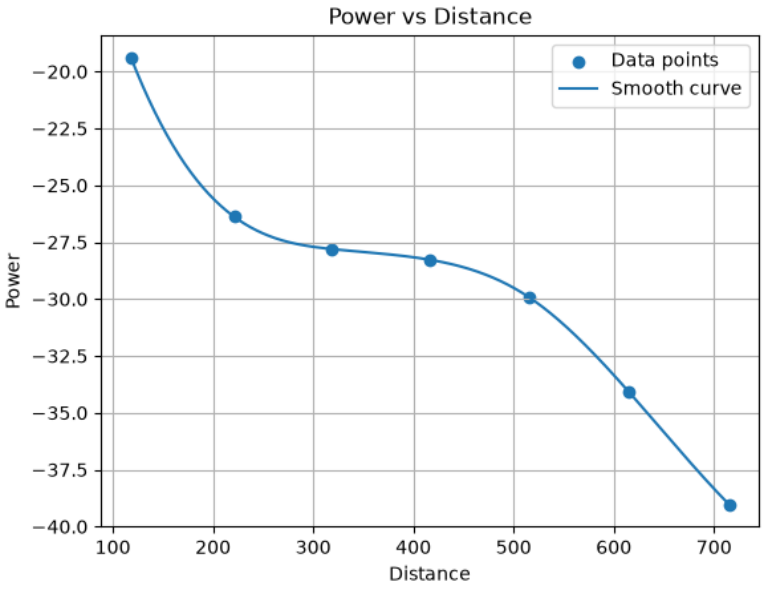
Part A.2 : VSWR and Return Loss :

Due to the constraints , the VSWR part were not performed .

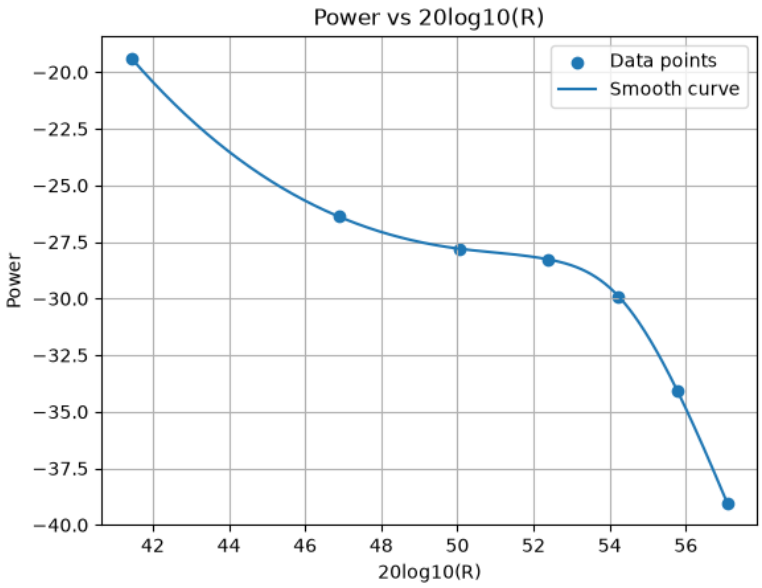
1.6.2 Part B: Effect of distance

1. Connect the antennas as shown .
2. Set a suitable frequency and amplitude in the signal generator within the operating range of the antenna.
3. Adjust the length of the monopole according to the selected frequency.
4. Keep the transmitting and receiving antennas aligned with the same polarization and at the same height.
5. Vary the distance between the transmitting and receiving antennas and record the received power.
6. Plot the received power P_r versus distance R.
7. Plot P_r versus $20\log_{10} R$ and comment on the observed trend.

PLOT OF RECEIVED POWER P_r VERSUS DISTANCE R



PLOT OF POWER VS 20LOG10 R :



The received power was measured by varying the antenna angle θ from 0° to 340° in steps of 20°, while keeping the other experimental parameters approximately constant. The measured received power varied between −46.87 dBm and −58.21 dBm, with an average value of approximately −52.43 dBm.

The received power does not remain constant with angle and shows noticeable fluctuations as the antenna is rotated. The maximum measured power of −46.87 dBm was obtained at approximately 280°, while the minimum power of −58.21 dBm was observed at approximately 300°. Thus, the total variation in received power over the angular range was about 11.34 dB.

The variation in received power with angle demonstrates the directional response of the antenna. The fluctuations can also be affected by antenna alignment, reflections from surrounding objects, multipath propagation, and experimental measurement uncertainties.

1.6.3 Part C: Beam pattern – co-polarization measurement

Keep the distance between the transmitting and receiving antennas at a minimum of 3λ. Align the electric fields of the transmitting and receiving antennas in the same plane and keep the antennas at the same height.

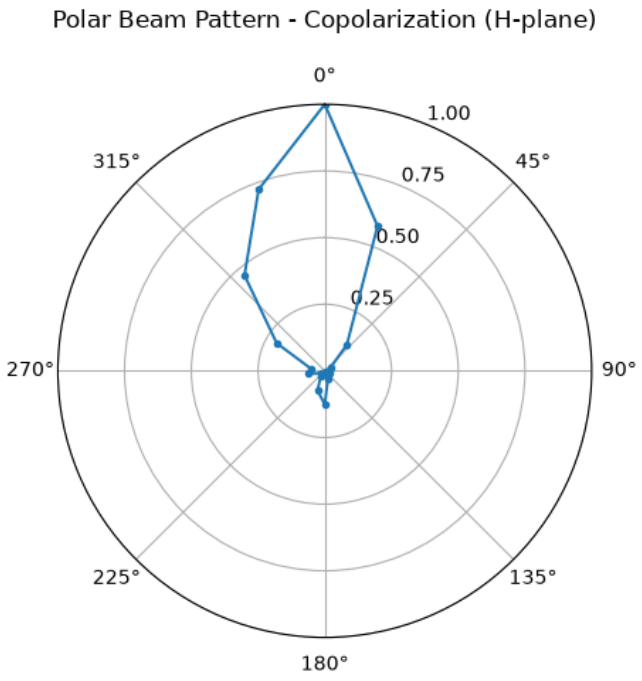
1. Set a fixed transmitting frequency within the operating range of the antenna.
2. Vary the E-plane angle θ from -90° to $+90^{\circ}$ and record the received power at suitable angular intervals.
3. Return the antenna to the reference position.

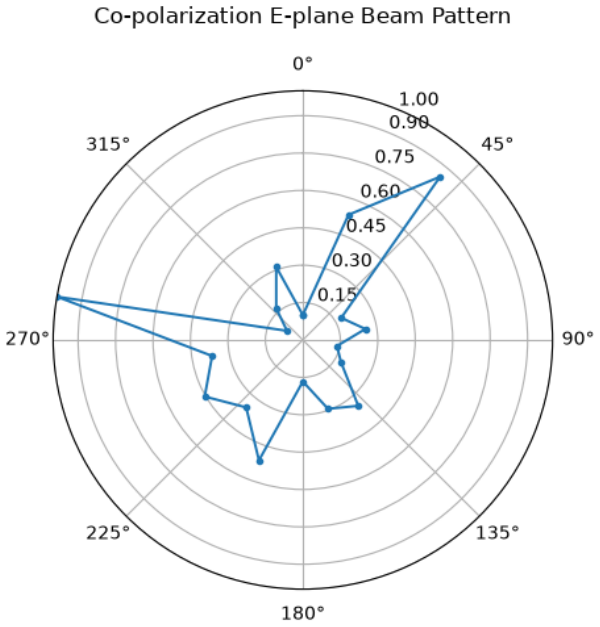
4. Vary the H-plane angle ϕ from -90° to $+90^\circ$ and record the received power.

5. Plot the E-plane and H-plane beam patterns as polar plots.

6. Repeat the experiment for two additional frequency values and compare the plots.

Angle(degree)	Co-polarization (dBm)		Cross-polarization (dBm)	
	E-Plane	H-plane	E-plane	H-plane
0	-56.87	-28.5	-51.75	-28.52
20	-49.59	-30.88	-44.5	-30.5
40	-47.55	-37.39	-41	-34.2
60	-54.38	-44.46	-43.51	-36.7
80	-52.82	-46.17	-49.45	-42.3
100	-55.44	-52.92	-49.55	-45.05
120	-54.42	-45.48	-47.6	-47.12
140	-51.49	-45.45	-53.95	-45.75
160	-52.19	-43.24	-53.15	-42.3
180	-54.65	-37.49	-52.15	-45.335
200	-49.76	-39.55	-46.56	-45.15
220	-51.39	-44.88	-52.5	-51.4
240	-50.3	-45.7	-47.6	-48.07
260	-51.2	-40.47	-54	-48.8
280	-46.87	-41.17	-48.9	-43.4
300	-58.21	-35.32	-45.7	-38.8
320	-54.67	-31.8	-48.41	-33.35
340	-51.93	-29.9	-50.6	-34.7
360	-56.87	-28.5	-51.75	-28.52





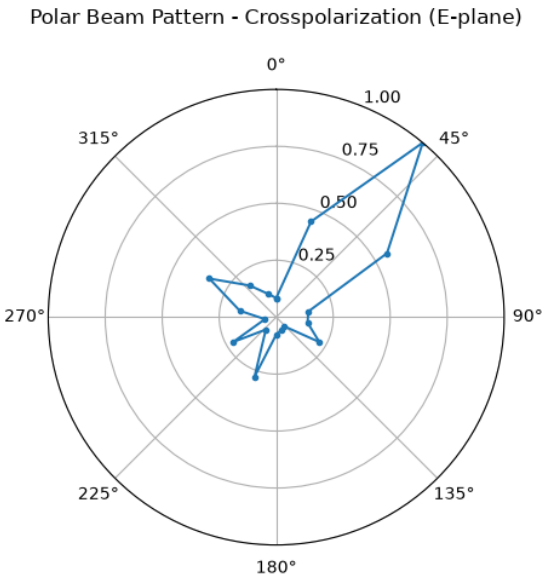
1.6.4 Part D: Beam pattern – cross-polarization measurement

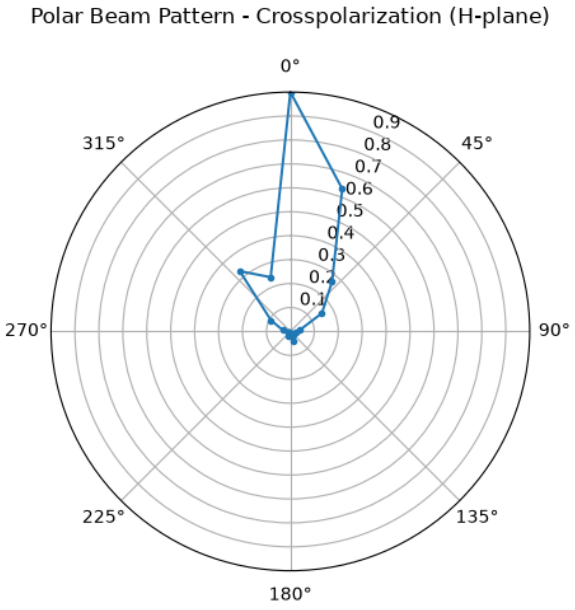
Adjust the transmitting antenna so that the electric fields of the transmitting and receiving antennas are perpendicular.

1. Retain the same frequency, antenna separation, heights, angular steps, and instrument settings used for the co-polarization measurement.

2. Vary the E-plane angle θ and record the received power.

3. Vary the H-plane angle ϕ and record the received power.
4. Plot the E-plane and H-plane beam patterns for the cross-polarization configuration.
5. Compare the co-polarization and cross-polarization measurements.





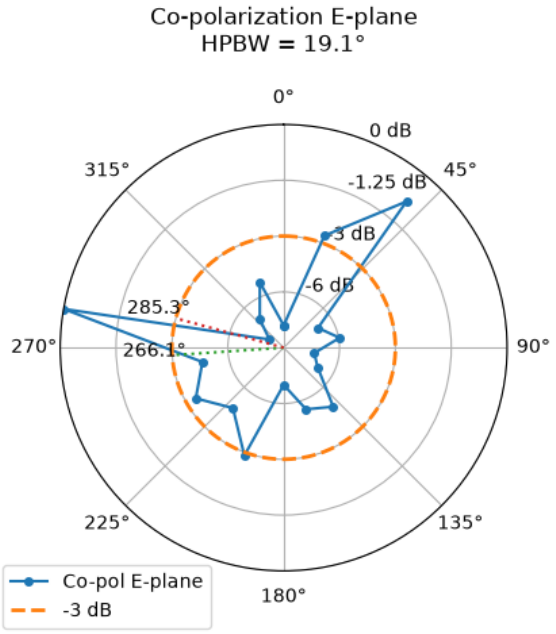
1.6.5 Part E: Half-power beamwidth and directivity

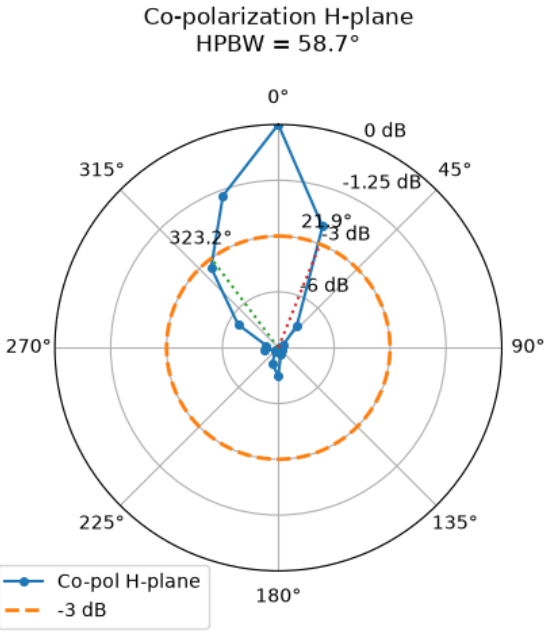
1. From the co-polarized E-plane beam pattern, identify the two -3 dB points and calculate the E-plane HPBW θ_E.

2. From the co-polarized H-plane beam pattern, identify the two -3 dB points and calculate the H-plane HPBW θ_H.

3. Convert θ_E and θ_H to radians and estimate the directivity using

$$D \approx \frac{4\pi}{\theta_E \theta_H.}$$





E-plane : HPBW θ_E = 19.1 degree (0.3333 rad)

H-plane : : HPBW θ_H = 58.7 degree (1.024 rad)

$$D \approx \frac{4\pi}{\theta_E \theta_H.}$$ D = 37.33

The absolute directivity ratio to dB gives estimated directivity = 15.72dB.

1.7 Results and Observations

Summarize the measured and calculated quantities in the following table.

Quantity	Symbol	Measured/Calculated Value
Antenna operating frequency (MHz)	f_r	1.5GHz
E-plane half-power beamwidth	θ_E	19.1 degree (0.333 rad)

H-plane half-power beamwidth	θ_H	58.7 degree (1.024 rad)
Estimated directivity	D	15.72dB

Results included in this report mentioned above :

1.8 Conclusions

State the following results and briefly discuss their physical significance:

- **measured antenna operating frequency :**

  The theoretical operating frequency for the 5 cm quarter-wave monopole was calculated 1.5GHz.

- **variation of received power with distance:**

  The received power has been decreased with distance as can be seen by the plots of Power vs Distance and Power vs Power vs 20log10(R). It is experimentally shown that antenna's path loss strictly follows the inverse-square law.

- **shapes of the E-plane and H-plane beam patterns and comparison of co-polarization and cross-polarization measurements:**

  The co-polarized angular measurements show directional beam patterns with distinct maxima and minima. The E-plane and H-plane patterns are different, demonstrating that antenna response depends on the plane of measurement. In cross-polarization, the transmitting and receiving electric fields are perpendicular, so the ideally received component would be strongly suppressed. The measured non-zero cross-polar response is expected in a real laboratory setup because of imperfect polarization purity, alignment errors, reflections and multipath, and finite antenna cross-polar response.

- **measured half-power beamwidths and estimated directivity** :

  As extracted from above θ_E = 19.1 degree(0.333 rad) and θ_H = 58.7 degree (1.02 rad) and estimated directivity D = 15.72 dBm . It clearly indicates that a highly directional radiated beam.

- **variation of the beam pattern with frequency:**

  The measured co-polarization patterns show the angular response of the antenna at that selected frequency; to study frequency dependence, the same angular measurements would need to be repeated at different frequencies.

- **Measurement Uncertainty :**

- Small errors in angular positioning of the antenna.
- Imperfect alignment of the transmitting and receiving antennas.
- Reflections from nearby objects, laboratory walls and equipment, producing multipath interference.
- Finite cross-polarization response of the antennas.
- Spectrum-analyser measurement uncertainty and fluctuations in the received signal.
- Changes in cable positioning or connector losses.
- The 20° angular sampling interval limits the precision with which beam maxima and −3 dB crossing points can be located.